
Multiple Choice Questions

1 point MCQs

Question 1 (1 point)

Let L be a language accepted by a DFA with q states. Which of the following statements is necessarily true?

- A. There exists a NFA with q or less states which accepts L .**
- B. Every NFA which accepts L requires atleast 2^q states.
- C. The number of strings in L is $q - 1$.
- D. The number of strings in L is $q + 1$.

Question 2 (1 point)

Let $L \subseteq \{0, 1\}^*$. Which of the following is true?

- A. If L is regular, all subsets of L are regular.
- B. If all finite subsets of L are regular, then L is regular.
- C. If a proper subset of L is not regular, then L is not regular.
- D. If all proper subsets of L are regular, then L is regular.**

Question 3 (1 point)

Let $L \subseteq \{0, 1\}^*$ and F be a finite subset of $\{0, 1\}^*$. Which of the following statements is true?

- A. $L - F$ is regular.
- B. $L \cap F$ is regular.**
- C. $L \cap F$ need not be regular.
- D. $L \cup F$ is regular.

Question 4 (1 point)

Let P, Q and R be regular expressions such that the number of strings generated by P is p , Q is q and R is r . What is the number of strings generated by the regular expression $(P + R)^*Q + PQ$?

- A. $(p + r)q + pq$
- B. $2^{(p+r)}q + pq$
- C. $(2^{pq} + q)(p + q)$
- D. Infinite.**

Question 5 (1 point)

Let $L = \{w \mid w \text{ has even length and odd number of 0's}\}$. Which of the following words is in L^* ?

- A. 0000.
- B. 0101010.
- C. 111101.**
- D. 010.

Question 6 (1 point)

What is the language corresponding to the following regular expression?

$(0 + 1)^*(01 + 10)(0 + 1)^*$

- A. $\{w \mid w \text{ contains } 01 \text{ or } 10 \text{ as a substring}\}$.**
- B. $\{w \mid \text{length of } w \text{ is } \geq 2\}$.

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- C. $\{ w \mid w \in \{0,1\}^* \}$.
- D. $\{ w \mid w \text{ contains atleast two 0s and two 1s} \}$.

Question 7 (1 point)

$L_1 = \{w \in \{0,1\}^* \mid w \text{ has at least as many occurrences of } (110)\text{'s as } (011)\text{'s} \}$.

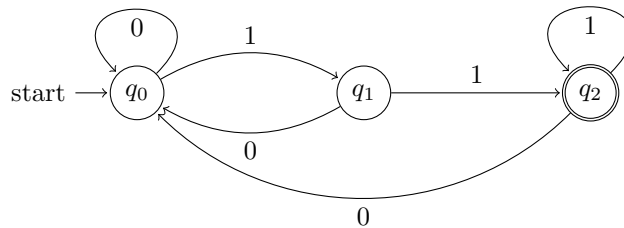
$L_2 = \{w \in \{0,1\}^* \mid w \text{ has at least as many occurrences of } (000)\text{'s as } (111)\text{'s} \}$.

Which of the following is TRUE?

- A. L_2 is regular but L_1 is not.
- B. L_1 and L_2 both are regular.
- C. L_1 is regular but L_2 is not.**
- D. Neither L_1 nor L_2 is regular.

Question 8 (1 point)

Which of the following is the correct description of language recognized by the DFA below?



- A. Set of binary strings with at least two 1s.
- B. Set of binary strings which ends with 11.**
- C. Set of binary strings which contains 11 as substring.
- D. Set of binary strings which has even number of 1s.